papergrid

Date: / /

Module 1

04/01/26 Computer security - By NIST

The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability and confidentiality of information system resources including hardware, software, firmware, information or data.

C - confidentiality - only sender & intended ~~replacement~~ recipients have access

I - Integrity - content of the msg can't be altered ~~after~~ sending to n recipient

A - Availability - resources should be available to authorised parties at all time

Accountability / Non-repudiation - an entity should be responsible for the message

Authenticity - proving the identity of the person who is accessing the information.

Access control - assign roles to user, and control / manage their access.

→ All these together form the principles of security

* Vulnerability - a weakness that a threat can import
* Threat - potential danger that can harm an asset
* Attack - actual intentional action taken to exploit the vulnerability for malicious actions.

Interruption: Another party trying to block or disrupt a msg transfer so the receiver is unable to receive the msg.

A → ✗ → B
C

Interception: Another party silently/secretly gaining access to the msg being transferred ie receiver receives the msg but ~~as~~ a third party also get the access to the info

A → B
↑ C

Modification: the message from sender is modified by third party and the receiver will receive msg but it's from third party and the receiver is unable to know.

A → B
↓ C

Fabrication - False info is given to receiver by third party.

A → B → C adds on some other facts into msg & then sends it to B as if the msg were coming from A but A didn't send any msg to B.

Types of attack

1. Active attack - An attack where data is accessed, as well as damage has been done in the process.
   eg. Interruption, Fabrication, Modification.

2. Passive attack - An attack where some data is accessed by the attacker, but no damage has been done in the process
   eg: Interception.

Types of active and passive attacks and which principle is being affected.

1) Active Attacks
These attacks directly interfere with system resources, modifying data or disrupting services, leading to data loss, integrity issues or denial of access.
* principle affected: Integrity and availability

~~Kinds~~ Types of active attacks

1. Masquerade: Attacker impersonates a legitimate user or system to gain unauthorized access
   eg: using stolen credentials to log in
   Attack type: Fabrication
   Security model violated or affected principles: Authenticity, integrity
   Reason: S/m fails to correctly verify the identity of the user.

2. Replay Attack: capturing and re-sending valid data.
   eg: Reusing authentication packets.
   Attack type: Fabrication
   affected principle: authenticity, integrity.
   Reason: old msg are accepted as new, valid communications

3. Message Modification : attacker ~~attends~~ alters the msg during transmission
eg: changing bank transaction amounts.
Attack type: Modification
affected principles: Integrity, authenticity.
Reason: the contents of the msg are changed without authorization.

4. Denial of service (DOS) : attaker prevents legitimate users from accessing services.
eg: Flooding a server with requests.
Affected principle: Availability
Reason: legitimate users cannot access sim resources.

2) Passive attack
It involve monitoring or observing the system without modifying data or disrupting services
* principle affected - confidentiality

Types
1. Release of message contents : attacker reads confidential data such as email, passwords or files
eg: Reading emails, listening to phone calls, packet sniffing.
Affected principle - confidentiality
Attack type - Interruption
Reason - Unauthorized access to sensitive information.

2. Traffic Analysis : Attacker observes communication patterns (frequency, size, timing).
eg: Identifying who communicates with whom and when
Affected principle - confidentiality.
Attack type : Interception
Reason - even w/o reading data, useful info is leaked.

13/01/26 Cryptography
Cryptography is the art of achieving security by encoding messages to make them non readable.

Cryptanalysis
It is the technique of decoding messages from a non readable format back to a readable format w/o knowing how they were initialy converted from readable to non readable

* Cryptology is the combination of cryptography and cryptanalysis

- Plain text - message readable by sender, recipient and anyone who has access to it.
- Cipher text - encoded plain text

Sender
PT
↓
encryption
↓
CT — transmission

Receiver
PT
↑
decryption
↑
CT

Cipher

substitutional cipher
alphabets or chars will be replaced/substituted using other chars

transposition cipher
alphabets or chars are rearranged to form transposition cipher.

Caesar cipher
choose alphabets that are 3 positions away
A→z ie 0→25
PT→A , CT→D A is coded to D
$y = E(x) = (x+3) \bmod 26$
modified ~~ca~~ caesar cipher (instead of 3 → n)
$y = (x+n) \bmod 26$

eg: H E L L O — plain text
7 4 11 11 14
10 7 14 14 17
K H O O R — cipher text

eg: $E(x) = (x+n) \bmod 26$ $n=5$

G O O D M O R N I N G — PT
6 14 14 3 12 14 17 13 8 13 6
11 19 19 8 17 19 22 18 13 18 11
L T T I R T W S N S L — CT

Affine cipher
Encryption is done using the eqn:
$E(x) = (ax+b) \bmod 26$ where a and b are key values.
decryption is done using the eqn:
$D(y) = a^{-1}(y-b) \bmod 26$ a and 26 should be relatively prime (co-prime)
here, a should be integer such that a^{-1} exist.

eg: plain text - MATH. perform affine cipher using $a=3$, $b=7$

M A T H
12 0 19 7
43 7 64 28 % 26
17 7 12 2
R H M C

14/01/26 Encrypt the plain text CODE. $a=5$ and $b=8$.

C O D E
2 14 3 4
18 78 23 28 % 26
18 0 23 2
S A X C

decryption.
$D(y) = a^{-1}(y-b) \bmod 26$

S A X C
18 0 23 2

To find a^{-1}

$a \times a^{-1} \bmod 26 = 1$

$5 \times a^{-1} \bmod 26 = 1$

$a^{-1} = 21$ (trial & error method)

$21(y-b) \% 26$

S A X C
18 0 23 2
210 168 315 126 % 26
2 14 9 4

Q. do decryption J D G A, a=7 and b=3

Ans: $D(y) = a^{-1}(y-b) \% 26$

$7 \times a^{-1} \bmod 26 = 1$

$a^{-1} = 15$

$15(y-3) \% 26$

J D G A
9 3 6 0
90 0 45 -45 % 26.
12 0 19 7
M A T H

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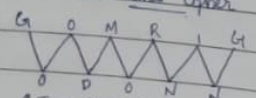
Rail Fence cipher



CT → GOMRIGODONN

PT → GOOD MORNING

here depth or no. of rails = 2
here 2 means make a grid with 2 row and fill it diagonally.

depth / no. of rails = 3

G				M				I		
	O		D		O		N		N	
		O				R				G

CT → GMIODONNORG

Q. PT → CRYPTOGRAPHY depth = 3

C				T				A		
	R		P		O		R		P		Y
		Y				G				H	

CT → CTARPORPYYGH

Columnar cipher / Columnar transposition cipher
It involves writing plain text in row and reading cipher text in column.

eg: PT → ATTACK TODAY
keyword → HACK

Ans: HACK
3 1 2 4 → alphabetic order

3	1	2	4
A	T	T	A
C	K	T	O
D	A	Y	X

CT → TKATTYACDAOX

Eg: PT → THIS IS A SECRET

keyword - KEY
2 1 3

2	1	3
T	H	I
S	I	S
A	S	E
C	R	E
T	X	X

CT → HISRXTSACTISEEX

Eg: PT → BATTLE OF BERLIN

keyword - TANK
4 1 3 2

4	1	3	2
B	A	T	T
L	E	O	F
B	E	R	L
I	N	X	X

CT → AEENTFLXTORX BLBI

Decrypt the cipher text using columnar cipher

CT → EVLNACDTMESEAROFO DEEC WIREE
E N O CDEECX

Keyword - ZEBRAS
6 3 2 4 1 5

6	3	2	4	1	5
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	O	N	C	E	X

cipher text has 30 alpha
so, no. of rows = 5.

PT → WE ARE DISCOVERED AT ~~FLE~~ FLEE AT ONCE ONCE

Playfair cipher

→ I and J are taken to be I itself inorder to accomodate all 26 alphabets in 5x5 matrix or grid.

Q. Create a grid using MONARCHY as keyword.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

plain text - INSTRUMENTS

Encryption rules:

* Split the PT into pairs of letters or digraph
* if PT has odd number of letters, append 'z' at end
* for any letter that appears twize in digraph place an X there.

To identify the cipher text create a 5x5 key matrix after that following 3 scenarios can occur

→ if 2 letters ~~should~~ occur together in a row as a digraph, each letter should be replaced with the letter that is immediate to its right.

→ if pair of letters in the digraph appear in the same column, each letter is replaced with letters immediately below them, if there is no letter below, replace with the top letter in the column.

→ Pair of letter is present in corresponding rows and columns consider a subset with that letters as diagonal elements & replace with letters from opposite elements.

Applying the rule.

INSTRUMENTS

IN → GA

ST → TL CT → GATLMZCLRQTX

RU → MZ

ME → CL

NT → RQ

SZ → TX

Q. using the same keyword ~~decrypt~~ encrypt the PT BALLOON

BA - IB ↳ BALXLOON

LX - SU CT → IBSUPMNA

LO - PM

ON -

Q. PT → DEFEND THE EAT WALL

Keyword - GUIDANCE

G	U	I	D	A
N	C	E	B	F
H	K	L	M	O
P	Q	R	S	T
V	W	X	Y	Z

DEFENDTHEXEASTWALXLZ

DE - IB

FE - NB

ND - BG

TH - PO

EX - LI

EA - FI

ST - TP

WA - ZU

LX - RI

LZ - DX

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Hill cipher

- polyalphabetic block cipher.
- It uses linear algebra to encrypt plain text by substituting blocks of letters.
- It offers better resistance to frequency analysis.

$C = KP \bmod 26$

where K → key matrix and P → Plain text

$P = K^{-1}C \bmod 26$ → decryption

eg. PT = attack $K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}_{2\times2}$

Since key matrix is 2x2 matrix, then we need to divide PT into block of 2 and follow column matrix.

$\begin{bmatrix} a \\ t \end{bmatrix} \begin{bmatrix} t \\ a \end{bmatrix} \begin{bmatrix} c \\ k \end{bmatrix}$ then convert plain text vectors into numerical equivalence

$\begin{bmatrix} 0 \\ 19 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

1st vector

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \bmod 26 = \begin{bmatrix} 57 \\ 114 \end{bmatrix} \bmod 26 = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} F \\ K \end{bmatrix}$

2nd vector

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \bmod 26 = \begin{bmatrix} 38 \\ 57 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 5 \end{bmatrix} = \begin{bmatrix} M \\ F \end{bmatrix}$

3rd vector

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \bmod 26 = \begin{bmatrix} 34 \\ 66 \end{bmatrix} \bmod 26 = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} I \\ O \end{bmatrix}$

CT → FKMFIO

$K^{-1} = \frac{1}{|K|} \text{adj } K$ $D = |K| = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} = 2\times6 - 3\times3$
$= 12 - 9 = 3$

$DD^{-1} = 1 \bmod 26$

$3D^{-1} = 1 \bmod 26$

$3D^{-1} \bmod 26 = 1$ $\therefore D^{-1} = 9$

$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ adj $K = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

adj $K = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix}$

$K^{-1} = D^{-1}(\text{adj } K) = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} 9 = \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$

CT

$\begin{bmatrix} F \\ K \end{bmatrix} \begin{bmatrix} M \\ F \end{bmatrix} \begin{bmatrix} I \\ O \end{bmatrix} \rightarrow \begin{bmatrix} 5 \\ 10 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \begin{bmatrix} 8 \\ 14 \end{bmatrix}$

1st vector

$P = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \bmod 26 = \begin{bmatrix} 260 \\ 305 \end{bmatrix} \bmod 26 = \begin{bmatrix} 0 \\ 19 \end{bmatrix}$

2nd vector

$P = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \bmod 26 = \begin{bmatrix} 149 \\ 390 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$

3rd vector

$P = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 8 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} 366 \\ 452 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

1. using hill cipher encrypt the plain text → Act where $K = \begin{bmatrix} 6 & 24 \\ 1 & 38 \end{bmatrix}$

2. Find the inverse and decrypt the CT → ZQ given that $K = \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix}$

Ans: 1. PT = ACT $K = \begin{bmatrix} 6 & 24 \\ 1 & 38 \end{bmatrix}$

$C = KP \bmod 26$

$P = \begin{bmatrix} A \\ C \end{bmatrix} \begin{bmatrix} T \\ X \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \begin{bmatrix} 19 \\ 23 \end{bmatrix}$

1st vector

$C = \begin{bmatrix} 6 & 24 \\ 1 & 38 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \bmod 26 = \begin{bmatrix} 48 \\ 76 \end{bmatrix} \bmod 26 = \begin{bmatrix} 22 \\ 24 \end{bmatrix}$

2nd vector

$C = \begin{bmatrix} 6 & 24 \\ 1 & 38 \end{bmatrix} \begin{bmatrix} 19 \\ 23 \end{bmatrix} \bmod 26 = \begin{bmatrix} 666 \\ 893 \end{bmatrix} \bmod 26 = \begin{bmatrix} 16 \\ 9 \end{bmatrix}$

$C = \begin{bmatrix} 22 \\ 24 \end{bmatrix} \begin{bmatrix} 16 \\ 9 \end{bmatrix}$

$C = \begin{bmatrix} W \\ Y \end{bmatrix} \begin{bmatrix} Q \\ J \end{bmatrix}$

CT → WYQJ

Ans: 2. CT → ZQ $K = \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix}$ $C = \begin{bmatrix} Z \\ Q \end{bmatrix} = \begin{bmatrix} 25 \\ 16 \end{bmatrix}$

$P = K^{-1} C \bmod 26$

$K^{-1} = \frac{1}{D} \operatorname{adj} K = D^{-1} \operatorname{adj}(K)$

$D = \begin{vmatrix} 7 & 8 \\ 11 & 11 \end{vmatrix} = -11 \bmod 26$
$= 15$

$D \cdot D^{-1} \bmod 26 = 1$

$15 \cdot D^{-1} \bmod 26 = 1$

$D^{-1} = 7$

$\operatorname{adj} K = \begin{bmatrix} 11 & -8 \\ -11 & 7 \end{bmatrix} = \begin{bmatrix} 11 & 18 \\ 15 & 7 \end{bmatrix}$

$K^{-1} = D^{-1} \operatorname{adj} K = 15 \begin{bmatrix} 11 & 18 \\ 15 & 7 \end{bmatrix} = \begin{bmatrix} 165 & 270 \\ 225 & 105 \end{bmatrix} \bmod 26$
$= \begin{bmatrix} 9 & 10 \\ 17 & 1 \end{bmatrix}$

$CT \rightarrow \begin{bmatrix} Z \\ Q \end{bmatrix} = \begin{bmatrix} 25 \\ 16 \end{bmatrix}$

$P = \begin{bmatrix} 9 & 10 \\ 17 & 1 \end{bmatrix} \begin{bmatrix} 25 \\ 16 \end{bmatrix} \bmod 26 = \begin{bmatrix} 385 \\ 441 \end{bmatrix} \bmod 26 = \begin{bmatrix} 21 \\ 25 \end{bmatrix}$

02/02/26 **Cryptanalytic attacks**

1. Ciphertext-only Attack
2. Known plaintext Attack
3. Chosen Plaintext Attack
4. Chosen Ciphertext Attack
5. Chosen Text Attack

1. Cipher text only Attack
   - Attacker has access to the cipher text, but doesn't have

access on keys and decryption algorithm.
- most difficult form of attack
- They conduct frequency analysis and pattern recognition.

2 Known Plain text Attack
- Attacker has access to some plain text - cipher text pairs. Also access to CT, encryption algorithm.
- Based on the pairs, they judge the patterns.

3 Chosen Plaintext Attack
- Attacker has access to encryption system for an amount of time.
- They have independence of choosing the PT and they get the CT as output. But can't access to decryption algorithm.
- Here PT is chosen by the attacker.

4 Chosen Ciphertext Attack
- Attacker has access to the decryption system (but not the key or decryption algm)
- They get the decrypted PT as output.

5 Chosen Text Attack
- Combination of chosen PT and chosen CT attack.
- Attacker gains access to the transmission medium.

1) Difference b/w substitution and transposition cipher

Substitution cipher	Transposition cipher
* Replaces each letter with another letter or symbol	* doesn't replace letters, only changes their order.
* Actual letters are hidden	* Actual letters remain the same
* Eg: A→D, B→E etc.	* Eg: HELLO → OLLEH
* changes the meaning of characters	* changes only the position of characters
* Harder to guess w/o the key	* Easier to break using pattern analysis
* Focuses on changing symbols	* Focuses on rearranging symbols

2) Difference b/w monoalphabetic and polyalphabetic substitution

Monoalphabetic substitution	Polyalphabetic substitution
* uses one fixed alphabet for substitution	* uses multiple substitution alphabets
* Each letter is always replaced by the same letter	* A letter can be replaced by different letters
* Simple & easy to understand	* more complex & harder to understand
* Vulnerable to frequency analysis	* Resists frequency analysis better
* Eg: caesar cipher	* Eg: Vignere cipher

3) Difference b/w block and stream cipher

Block cipher	stream cipher
* Encrypts data in fixed size blocks	* Encrypts data one bit/byte at a time
* Same key is used for an entire block	* key stream changes continuously
* Slower for small amounts of data	* Faster and efficient for real time data

* Padding is required if data size is smaller than block size	* no padding is required
* Eg: AES	* Eg: RC

Vignere cipher

- A polyalphabetic substitution cipher
- uses different keys

Eg: PT → COFFEE
Key → CAKE

Make length of PT and key the same by repeating the key. then find the numerical value. add the numerical value of PT with the key and find the equivalent CT.

```
 2 14 5 5 4 4
 C O F F E E
 C A K E C A
 2 0 10 4 2 0
 ------------
 E O P J G E  → CT
 4 14 15 9 6 4
```

Encryption formula : (PT + Key) % 26
Decryption formula : (CT - Key) % 26

Q PT - Attack at dawn
Key - Lemon


```
 0 19 19 0 2 10  0 19  3 0 22 13
 A T T A C K  A T  D A W N
 L E M O N L  E M O N L E
 11 4 12 14 13 11 4 12 14 13 11 4
 ---------------------------------
 11 23 31 14 15 21 4 31 17 13 33 17   % 26
 11 23 5 14 15 21 4 5 17 13 7 17
 L X F O P V E F R N H R
```

Vernam cipher

- Also refers as OTP (one-time pad)
- either bit by bit or char by char
- Each bit ~~or char~~ of PT is XORed with random bit ~~or char~~ of key value result in CT

→ PT and key ~~same~~ must be of same length
character by character - modulo addition
bit by bit encryption - XOR

Eg: HI

```
 (H) 01001000  (I) 01001001        CT = PT ⊕ K
Key- 10110010   01101101
     11111010   00100100
```

To decrypt use CT ⊕ K = PT

If a normal word is to be encrypted bit by bit, then convert to ASCII and then find binary equivalent.

1. Eg: PT → CAT
Key → XMC

2 0 19
C A T
X M C
23 12 2
25 12 21
Z M V

2. Decrypt using Vernam cipher
CT → DJJTKYUPFYCV
Key → WJOPKLMNBVCX

3 9 9 19 10 24 20 15 5 24 2 21
D J J T K Y U P F Y C V –
W J O P K L M N B V C X
22 9 14 15 10 11 12 13 1 21 2 23
-19 0 -5 4 0 13 8 2 4 3 0 -2 % 26
7 0 21 4 0 13 8 2 4 3 0 24
H A V E A N I C E D A Y

PT → HAVE A NICE DAY

Mathematics of cryptography

Modular arithmetic
→ keeps keys & CT within a fixed-bit length

- often called clock arithmetic
- It deals with integers where nos wrap around upon reaching a certain value, called modulus (n).

→ $a \equiv b \pmod{n}$ means a & b have same remainder when divided by n.
↳ equivalent to saying that n divides the difference (a-b) perfectly.

Congruence
- It is the language of modular arithmetic.
- unlike std equality (=), a congruence (≡) describes a relationship b/w 2 nos within a specific modulus.

eg: $17 \equiv 5 \pmod{12}$
↳ on a std 12-hour clk, 17:00 (5 PM) is the same posⁿ as 5:00.

→ It allows us to work with mass nos by reducing them to a fixed range [0, n-1], keeping computations manageable for b/ds (cryptography use)

Euclidean Algorithm
→ check if nos are coprime, a req for key generation

Efficient method for finding Greatest common divisor (GCD) of 2 integers

Std Algm:
To find gcd(a,b):
1. divide a by b to get remainder r.

2. Replace a with b & b with r
3. Repeat until the remainder is 0. The last non-zero remainder is GCD

Extended Euclidean Algm (EEA) → actual tool used to compute private key from public key

- In cryptography, we use extended version. It not only finds the GCD but also finds integers x and y such that.

$ax + by = \gcd(a, b)$

↳ foundational step for calc private keys in RSA.

Modular Multiplicative Inverse → used in decryption to undo multiplication performed during encryption

- In modular arithmetic, we don't use fractions, we find an integer that acts like a fraction
- modular multiplicative inverse of a (p mod {n}) is an int x s:

$ax \equiv 1 \pmod n$

- inverse exists only iff a and n are coprime ($\gcd(a, n) = 1$).

Eg: inverse of 3 (mod 7)

$3 \times 5 = 15$

$15 \pmod 7 = 1$

∴ 5 is inverse of 3 (mod 7)

Problems

1. Find gcd (150, 35) using the Euclidean Algm.

Ans. $\frac{150}{35}$ remainder = 10 $\quad \frac{35}{10}$ remainder = 5

$\frac{10}{5}$ remainder = 0 $\quad$ GCD = 5 (non-zero remainder

2. Find gcd (a 210, b 192) using Euclidean Algm

Ans: a) ÷ a by b to get remainder r
b) replace a with b & b with r
c) repeat until remainder is 0 and last non-zero remainder is GCD.

$\frac{210}{192}$ remainder = 86 $\quad \frac{192}{86}$ remainder = 20

$\frac{86}{20}$ remainder = 6 $\quad \frac{20}{6}$ = remainder = 2

$\frac{6}{2}$ remainder = 0 $\quad$ ∴ GCD (210, 192) = 2

3. Calculate gcd (132, 54). Show all steps of the division.

Ans: a) $\frac{132}{54}$ remainder = 24

$$\begin{array}{r} 2 \\ 54\overline{)132} \\ 108 \\ \hline 24 \rightarrow r \end{array}$$

b) $\frac{54}{24}$ remainder = 6

$$\begin{array}{r} 2 \\ 24\overline{)54} \\ 48 \\ \hline 6 \end{array}$$

c) $\frac{24}{6}$ remainder = 0 $\quad$ ∴ gcd (132, 54) = 6

division steps
a) 132 = 2×54 + 24
b) 54 = 2×24 + 6
c) 24 = 3×6 + 0

4. Find the value of x such that 0 <= x < 13 for the congruence:

$45 \equiv x \pmod{13}$

Ans: 0 <= x < 13

$$\begin{array}{r} 3 \\ 13\overline{)45} \\ 39 \\ \hline 6 \rightarrow x \end{array}$$

congruence: $45 \equiv x \pmod{13}$

x is remainder when 45 is divided by 13

45 − 39 = 6

∴ x = 6

5. Simplify $(15 \times 12) \pmod 7$

$180 \bmod 7 = 5$

$$\begin{array}{r} 25 \\ 7 \overline{)180} \\ 14 \\ \hline 40 \\ 35 \\ \hline 5 \end{array}$$

6. Evaluate $(-10) \bmod 3$ ensure the result is a positive integer in the range $[0,2]$

Ans: using quotient - remainder thm,

$a \pmod n$ is defined by eqn $a = nq + r$, where q an integer quotient and r is remainder such that $0 \le r < n$.

to ensure r is non-negative, need to find largest q s.t. $nq \le a$.

$\therefore$ $-10 \div 3$ gives -3.33 as q so largest int less than or equal to this value is $q = -4$

$\therefore$ $a = -10$, $n = 3$ and $q = -4$

$a = nq + r$

$r = a - nq = -10 - 3 \times -4 = -10 + 12$

$r = 2$

$0 \le 2 < 3$ $\therefore$ it comes under the range $[0,2]$.

$-10 \bmod 3 = 2$

7. Given $\gcd(42, 15) = 3$, find integers x and y such that $42x + 15y = 3$

Ans: 1) find $\gcd(42,15)$ using euclidean algm through series of division

a) $42 = 2 \times 15 + 12$

b) $15 = 1 \times 12 + 3$

c) $12 = 4 \times 3 + 0$

$\therefore$ $\gcd(42, 15) = 3$

2) Extended Euclidean Algm (Back - substitution)

To find x & y, perform back - substitution using eqn from 1)

from 2nd eqn : $3 = 15 - 1 \times 12$

from 1st eqn : $12 = 42 - 2 \times 15$

Sub eqn for 12 into eqn for 3:

$3 = 15 - 1 \times (42 - 2 \times 15)$

$3 = 15 - 1 \times 42 + 2 \times 15$

$3 = -1 \times 42 + 3 \times 15$

$\rightarrow$ $42(-1) + 15(3) = 3$

Compare with $42x + 15y = 3$

we get $x = -1$ and $y = 3$

8. Find x and y such that $\gcd(81, 57) = 81x + 57y$

Ans: 1) Find $\gcd(81, 57)$ using euclidean algm through series of division.

a) $81 = 1 \times 57 + 24$

b) $57 = 2 \times 24 + 9$

c) $24 = 2 \times 9 + 6$

d) $9 = 1 \times 6 + 3$

e) $6 = 2 \times 3 + 0$ $\therefore$ $\gcd(81, 57) = 3$

2) Extended Euclidean algm (Back-substitution)

from d) : $3 = 9 - 1\times 6$

from c) : $6 = 24 - 2\times 9$

sub exp for 6 in d)

$3 = 9 - 1\times(24 - 2\times 9)$

$3 = 9 - 1\times 24 + 2\times 9$

$3 = 3\times 9 - 1\times 24$ — ①

from b) : $9 = 57 - 2\times 24$

from a) : $24 = 81 - 1\times 57$

Sub expn for 9 in ①

$3 = 3\times(57 - 2\times 24) - 1\times 24$

$3 = 3\times 57 - 6\times 24 - 1\times 24$

$3 = 3\times 57 - 7\times 24$ — ②

Sub expn for 24 in ②

$3 = 3\times 57 - 7\times(81 - 1\times 57)$

$= 3\times 57 - 7\times 81 + 7\times 57$

$3 = -7\times 81 + 10\times 57$

$3 = 81(-7) + 57(10)$

compare with $3 = 81x + 57y$

$\therefore x = -7$ and $y = 10$

9. For $a = 25$ and $b = 11$, find x and y that satisfy $25x + 11y = 1$

Ans: Given $25x + 11y = 1$ $\quad \gcd(25, 11) = 1$

1) Apply Euclidean Algm

1) $25 = 2\times 11 + 3$

2) $11 = 3\times 3 + 2$

3) $3 = 1\times 2 + 1$

4) $2 = 1\times 2 + 0$

$\therefore \gcd(25, 11) = 1$

2) Extended Euclidean Algm

from 3rd eqn : $1 = 3 - 1\times 2$

from 2nd eqn : $2 = 11 - 3\times 3$

from 1st eqn : $3 = 25 - 2\times 11$

sub expn of 2 in $1 = 3 - 1\times 2$

$1 = 3 - 1\times(11 - 3\times 3)$

$1 = 3 - 1\times 11 + 3\times 3$

$1 = 4\times 3 - 1\times 11$ — ①

sub expn of 3 in ①

$1 = 4\times(25 - 2\times 11) - 1\times 11$

$1 = 4\times 25 - 8\times 11 - 1\times 11$

$1 = 4\times 25 - 9\times 11$

$25(4) + 11(-9) = 1$

compare with $25x + 11y = 1$

$\therefore x = 4$ and $y = -9$

10. Find the modular inverse of 3 (mod 10) by testing small integers.

Ans: modular inverse of a (mod n) is an integer x such that

$ax \equiv 1 \pmod{n}$

here, $3x \equiv 1 \pmod{10}$

trying small integers

for $x=1 \rightarrow 3\times1=3 \equiv 3 \pmod{10}$

for $x=2 \rightarrow 3\times2=6 \equiv 6 \pmod{10}$

for $x=3 \rightarrow 3\times3=9 \equiv 9 \pmod{10}$

for $x=4 \rightarrow 3\times4=12 \equiv 2 \pmod{10}$

for $x=5 \rightarrow 3\times5=15 \equiv 5 \pmod{10}$

for $x=6 \rightarrow 3\times6=18 \equiv 8 \pmod{10}$

for $x=7 \rightarrow 3\times7=21 \equiv 1 \pmod{10}$

So from there $3\times7=21$ and $21=2\times10+1$

so the remainder is 1

$\therefore$ modular inverse of $3 \pmod{10}$ is 7

11 What is the multiplicate inverse of $5 \pmod{24}$

Ans: 1) Forward Euclidean Algm

1st we use the std Euclidean Algm to find the gcd (24, 5) and ensure it is 1.

A multiplicative inverse for $a \pmod n$ exists if and only if $\gcd(a,n)=1$.

1. $24 = 4\times5+4$
2. $5 = 1\times4+1$ → gcd is 1. inverse exists
3. $4 = 4\times1+0$

$\because$ gcd(24, 5) = 1, multiplicative inverse exists.

2) Extended Euclidean Algm (Back-substitution)

we now work backward from the eqn where the remainder is 1

1. we want to express 1 as a linear combination of 24 & 5

from 2nd eqn : $1 = 5 - 1\times4$ — ①

from 1st eqn : $4 = 24 - 4\times5$

Sub exp of 4 in ①

$1 = 5 - 1\times(24 - 4\times5)$

$1 = 5 - 1\times24 + 4\times5$

$1 = 5\times5 - 1\times24$

$\Rightarrow 1 = -1\times24 + 5\times5$

this eqn shows $5\cdot5 \equiv 1 \pmod{24}$

3) Verification

check the result by performing the multiplication:

$5\times5 = 25$

$25 \pmod{24} = 1$

because the product is congruent to $1 \pmod{24}$, 5 is its own multiplicative inverse

$\therefore$ Multiplicative inverse of $5 \pmod{24}$ is 5.

12. Find x such that $7x \equiv 1 \pmod{15}$

Ans: Forward Euclidean Algm

To find x in $7x \equiv 1 \pmod{15}$, need to check if sol^n exists by calc gcd (15, 7). A sol^n exists if gcd(15, 7) divides 1.

$\because$ 7 and 15 are coprime → gcd (15, 7) = 1 and because 1 divides 1, there is a unique sol^n for x modulo 15.

1. $15 = 2\times7+1$
2. $7 = 7\times1+0$

2) Extended Euclidean Algm

from 1st eqn: $1 = 15 - (2 \times 7)$

3) To Solve for x

the eqn $1 = 15(1) + 7(-2) \Rightarrow 7(-2) \equiv 1 \pmod{15}$.

$\therefore x \equiv -2 \pmod{15}$ is sol^n. Since we need +ve sol^n for x, add modulus 15 to it.

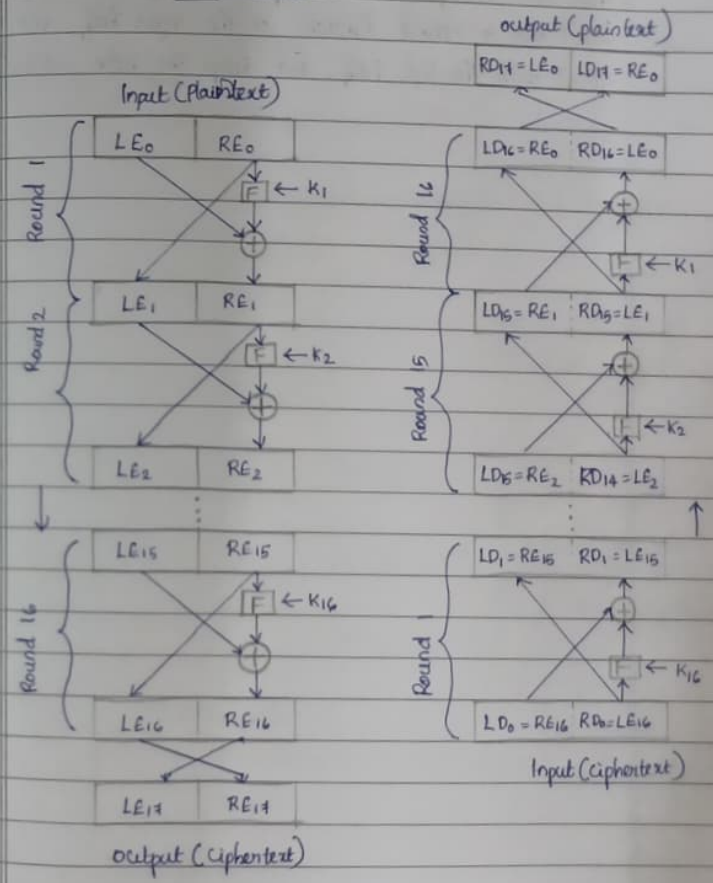
$x \equiv -2 + 15 \equiv 13 \pmod{15}$

Verification: $7 \times 13 = 91$. Since $91 = 6 \times 15 + 1$,

$\Rightarrow 91 \equiv 1 \pmod{15}$

So the value of x that satisfies the congruence is $x \equiv 13 \pmod{15}$.

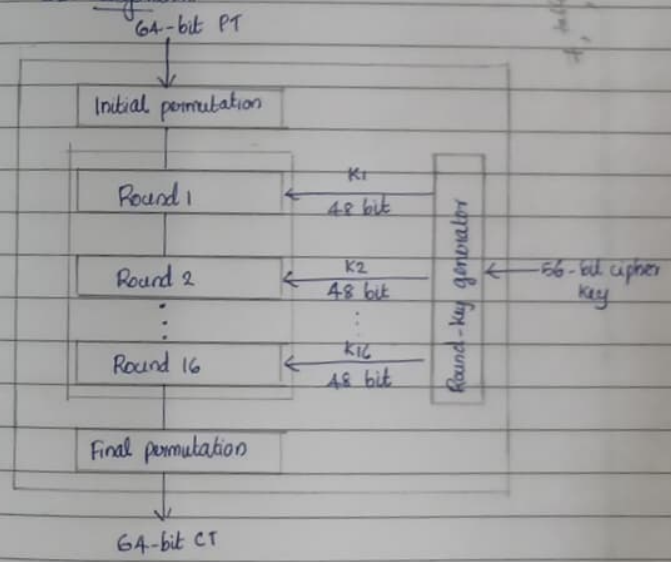
Feistel Structure



- A Feistel cipher is a symmetric block cipher structure that splits input data into left() and right() halves, processing

then over multiple rounds to provide high security. Each round uses a round function on the right half, XORs the result with the left half, and swaps the halves, allowing for

Data Encryption Standard (DES)

- Also called data encryption Algm (DEA). DES is a block cipher, it encrypts data in blocks of 64 bits each.
- The 64 bits of plain text goes as input to DES, which produces 64 bits of cipher text.
- The same algm & key are used for encryption and decryption.
- The key length is 56 bits

DES algorithm.